Discussion Exercise 8

University of California, Riverside
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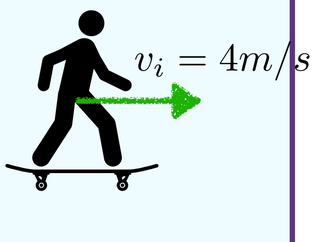
1. Conservation of Momentum

$$\vec{p} = m\vec{v}$$

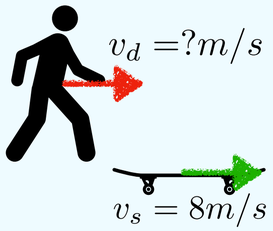
Dan is gliding on his skateboard at 4 m/s. He suddenly jumps backwards off the skateboard, kicking the skateboard forward at 8 m/s. How fast is Dan going as his feet hit the ground? Dan's mass is 50 kg and the skateboard has a mass of 5 kg.

Visualization

Initial state



Final state



Physics concepts (Conservation of momentum:)

$$\vec{p}_i = \vec{p}_f$$

$$p_i = (m_d + m_s)v_i$$

$$p_f = m_d v_d + m_s v_s$$

Known

$$m_d = 50kg$$

$$m_s = 5kg$$

$$v_i = 4m/s$$

$$v_s = 8m/s$$

Find

$$v_d = ?$$

$$\underbrace{(50+5)\times 4}_{p_i} = \underbrace{50v_d + 5\times 8}_{p_f}$$

$$v_d = 3.6(m/s)$$

Visualization → Physics concepts → Unknown $\xleftrightarrow{\text{Equations}}$ Known

Things you need to know

2. Conservation of Momentum (Collisions)

Types of collisions

Coefficient of restitution $(e) = \frac{|\text{Relative velocity after collision}|}{|\text{Relative velocity before collision}|}$

1. Perfectly elastic collsion $(e = 1)$
   - total momentum is conserved
   - total kinetic energy is conserved
2. Inelastic collsion ($0 \leq e < 1$, where $e = 0$ corresponds to perfectly inelastic collision)
   - total momentum is conserved
   - total kinetic energy is NOT conserved

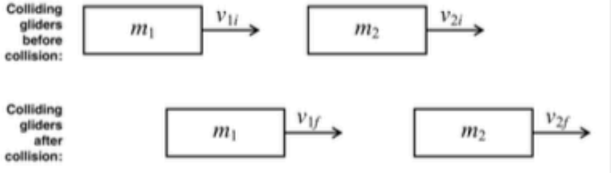
Things you need to know

1. Perfectly elastic collsion

- $m_1 v_{1i} + m_2 v_{2i} = m_1 v_{1f} + m_2 v_{2f}$
- $\frac{1}{2} m_1 v_{1i}^2 + \frac{1}{2} m_2 v_{2i}^2 = \frac{1}{2} m_1 v_{1f}^2 + \frac{1}{2} m_2 v_{2f}^2$

Predictions

$$v_{1f} = \frac{(m_1 - m_2) v_{1i} + 2 m_2 v_{2i}}{m_1 + m_2} \qquad v_{2f} = \frac{(m_2 - m_1) v_{2i} + 2 m_1 v_{1i}}{m_1 + m_2}$$

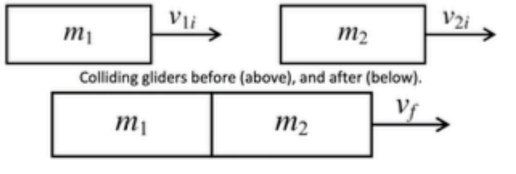


2. Perfectly inelastic collsion

- $m_1 v_{1i} + m_2 v_{2i} = (m_1 + m_2) v_f$
- $\frac{1}{2} m_1 v_{1i}^2 + \frac{1}{2} m_2 v_{2i}^2 > \frac{1}{2} (m_1 + m_2) v_f^2$

Prediction

$$v_f = \frac{m_1 v_{1i} + m_2 v_{2i}}{m_1 + m_2}$$



2. Conservation of Momentum

Two particles collide, one of which was initially moving and the other initially at rest.

1. Is it possible for *both* particles to be at rest after the collision? Give an example in which this happens, or explain why it can't happen.

If momentum of the system is conserved, then this cannot happen since initial momentum is finite while the final momentum vanishes

2. Is it possible for *one* particle to be at rest after the collision? Give an example in which this happens, or explain why it can't happen.

This happens when the momentum of the system is conserved and two particles have equal mass, and there is no energy loss during the collision (perfect elastic collision). As one can see from the equation on the RHS, if the mass are equal, two particles swap their velocities!

1. **Perfectly elastic collsion** ($e = 1$)

- $m_1 v_{1i} + m_2 v_{2i} = m_1 v_{1f} + m_2 v_{2f}$
- $\frac{1}{2} m_1 v_{1i}^2 + \frac{1}{2} m_2 v_{2i}^2 = \frac{1}{2} m_1 v_{1f}^2 + \frac{1}{2} m_2 v_{2f}^2$

Predictions

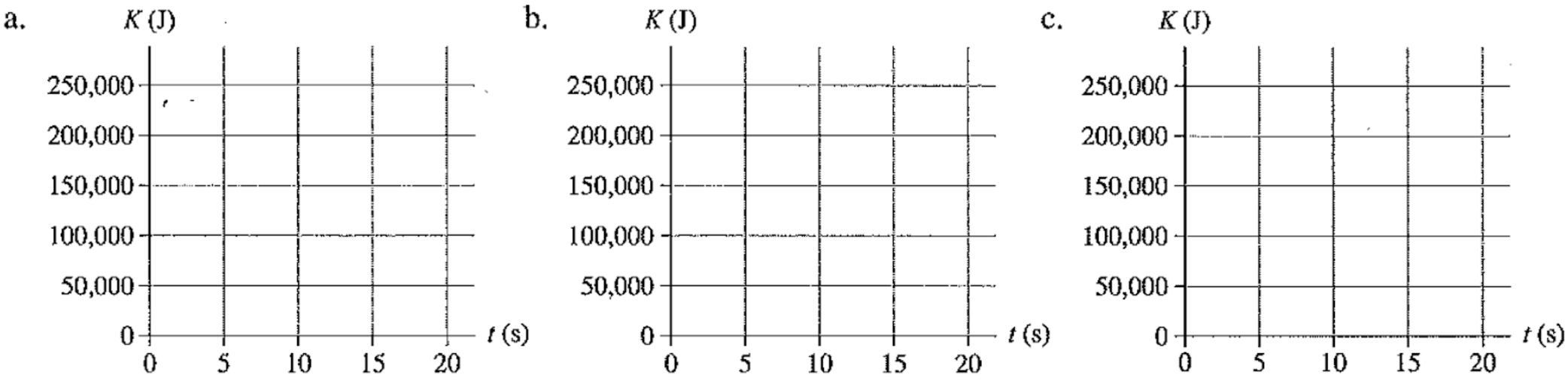
$v_{1f} = \frac{(m_1 - m_2) v_{1i} + 2 m_2 v_{2i}}{m_1 + m_2}$ $\quad v_{2f} = \frac{(m_2 - m_1) v_{2i} + 2 m_1 v_{1i}}{m_1 + m_2}$

3. Kinetic Energy

$$KE = \frac{1}{2}mv^2$$

Draw three sets of axes on the board as shown below. On each draw graphs of the kinetic energy of

a) a 1000 kg car that uniformly accelerates from 0 to 20 m/s in 20 s.
b) a 1000 kg car moving at 20 m/s that brakes to a halt with uniform decelaration in 20 s.
c) a 1000 kg car that drives once around a 130 m-diameter circle at a uniform speed of 20 m/s. *Mark several points, and then connect then by a smooth curve*



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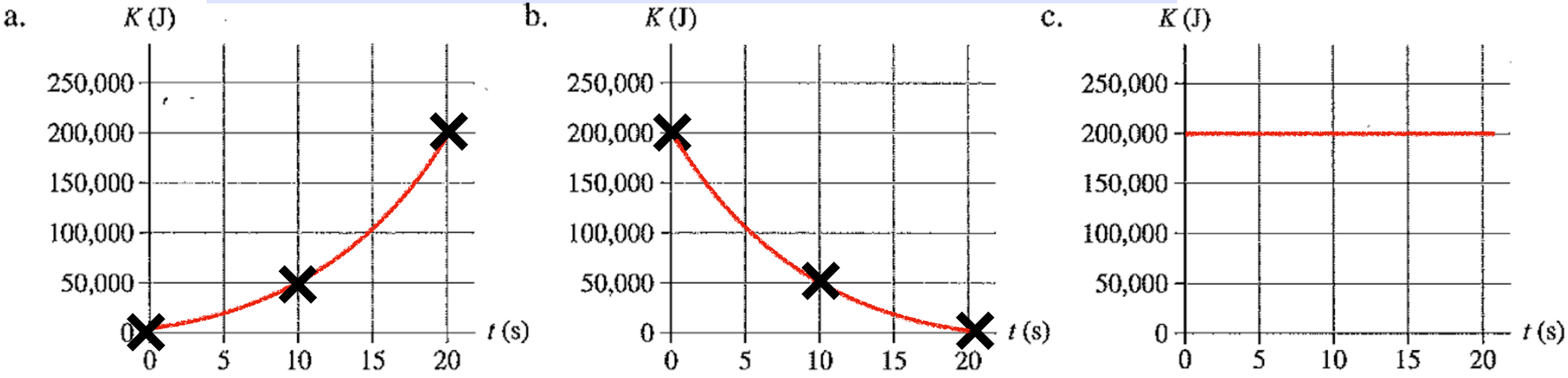


$$K_1 = \frac{1}{2}mv^2$$
$$= \frac{1}{2}m(at)^2 = 500t^2 \ (J)$$

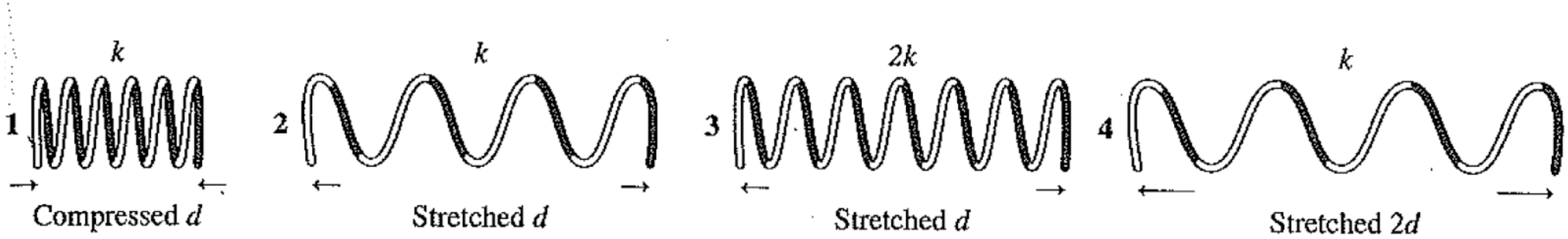
$$K_2 = \frac{1}{2}mv^2$$
$$= \frac{1}{2}m(v_i - at)^2 = 500(20-t)^2 \ (J)$$

$$K_3 = \frac{1}{2}mv^2$$
$$= \frac{1}{2} \times 1000 \times 20^2 (J) = 200000 (J)$$

4. Elastic Potential Energy

$$U_s = \frac{1}{2}kx^2$$

Rank in order, from most to least, the amount of elastic potential energy $(U_s)_1$ to $(U_s)_4$ stored in each of these springs. Explain your ordering.



$$(U_s)_1 = \frac{1}{2}kd^2 \qquad (U_s)_2 = \frac{1}{2}kd^2 \qquad (U_s)_2 = \frac{1}{2}(2k)d^2 = kd^2 \qquad (U_s)_4 = \frac{1}{2}(k)(2d)^2 = 2kd^2$$

$$(U_s)_4 > (U_s)_3 > (U_s)_1 = (U_s)_2$$